

ADP-DiT: Text-Guided Diffusion Transformer for Brain Image Generation in Alzheimer's Disease Progression

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github.com/labhai/ADP-DiT

1 Introduction

Problem. Alzheimer's disease progresses heterogeneously, making subject-specific modeling of longitudinal brain changes essential.

Goal. Given a baseline MRI and clinical metadata prompt, synthesize a subject-specific follow-up MRI at a target interval Δt .

Contributions

01 Interval-aware AD progression synthesis - generate follow-up MRI with Δt control using diagnosis, demographics, and 13 cognitive scores.

02 Clinical text-guided DiT - combine OpenCLIP and T5-XXL via cross-attention and AdaLN to guide subject-specific generation.

03 Anatomy-preserving latent DiT - use RoPE and SDXL-VAE latent diffusion to preserve spatial brain structure efficiently.

2 Dataset

712 participants **3,321** 3T T1w scans **259K** 2D slices

ADNI cohort; Each pair defines a **progression trajectory**:

Progression group	Subj.	Scans
CN → CN	160	755
CN → MCI	27	43
MCI → MCI	434	1,909
MCI → AD	117	234
AD → AD	106	380

Abbreviations: CN, Cognitively Normal; MCI, Mild Cognitive Impairment; AD, Alzheimer's Disease

Text Prompt Components

Diagnosis trajectory

CN **MCI** **AD**

Demographics + interval baseline → follow-up

Age **Sex** **Δt follow-up interval**

13 cognitive scores

Neuropsychological test results quantifying cognitive decline.

CDR-SB **MMSE** **MoCA** **ADAS-11** **ADAS-13**

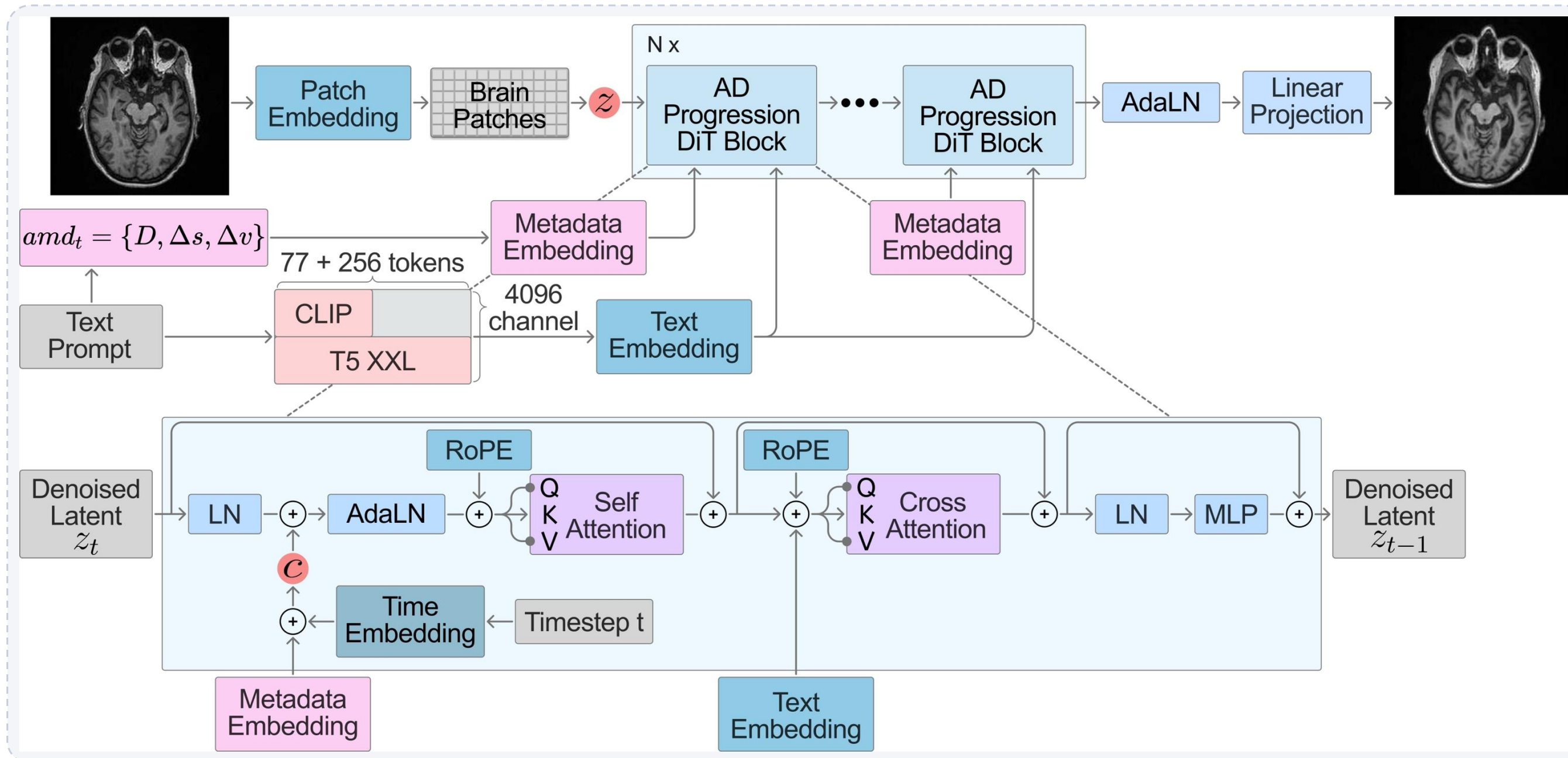
ADAS-Q4 **RAVLT-imm** **RAVLT-learn** **FAQ**

RAVLT-forget **RAVLT-%forget** **LDELTOTAL** **TRABSCOR**

Abbreviations: CDR-SB, Clinical Dementia Rating–Sum of Boxes; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; ADAS, Alzheimer's Disease Assessment Scale; RAVLT, Rey Auditory Verbal Learning Test; LDEL, Logical Memory Delayed Recall; TRAB, Trail Making Test Part B; FAQ, Functional Activities Questionnaire.

3 Method - ADP-DiT Architecture

ADP-DiT converts **follow-up interval Δt** , **demographics**, **diagnosis**, and **cognitive scores** into a clinical prompt, enabling time-specific and subject-specific AD MRI synthesis.



Clinical Prompt Encoding

Diagnosis, demographics, cognitive scores, and Δt are encoded as a single clinical prompt.

Dual Text Encoders

OpenCLIP captures vision–language semantics, while T5-XXL interprets detailed clinical context.

Hybrid Conditioning

Cross-attention injects fine-grained text guidance; AdaLN modulates global progression context.

RoPE Fusion — Spatially Consistent Latent DiT

RoPE encodes relative spatial relationships between MRI patches for anatomical consistency; SDXL-VAE enables efficient high-resolution synthesis in a compact latent space.

4 Quantitative

ADP-DiT achieves the highest SSIM and PSNR among generative baselines, outperforming DiT and text-guided models.

Model	SSIM $\uparrow$	PSNR $\uparrow$	MSE $\downarrow$
ADP-DiT	.8739	29.32	.0024
DiT (baseline)	.7652	23.24	.0052
SwinUNETR	.8535	28.37	.0019
Diffusion-CLIP	.6556	24.33	.0051
VQGAN-CLIP	.7463	22.97	.0055
Stable Diff. 2.1	.7952	23.44	.0055
FCDiffusion	.5474	20.55	.0106

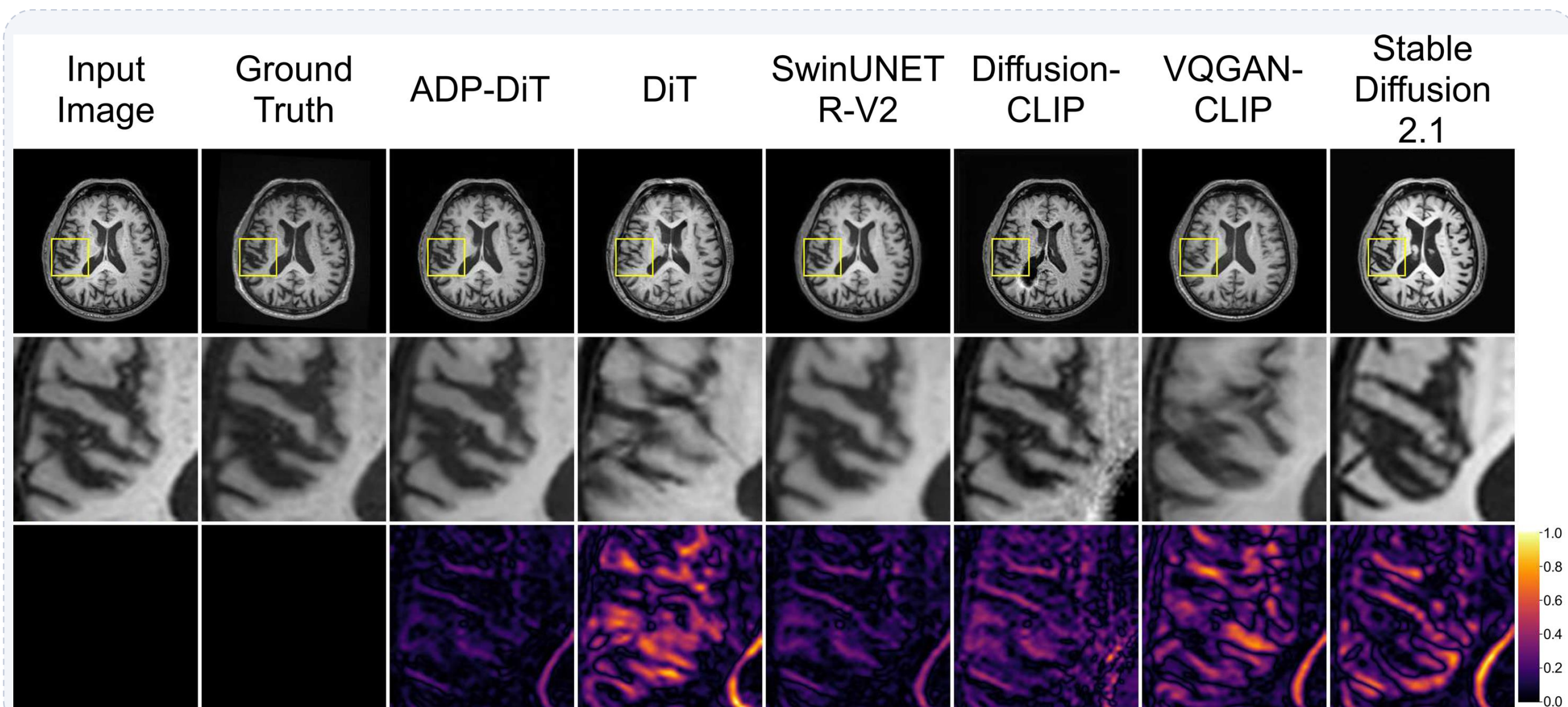
5 By Group & Δt

SSIM by Group	scale 0–1
CN → CN	0.8695
CN → MCI	0.8131
MCI → MCI	0.8779
MCI → AD	0.8227
AD → AD	0.8895

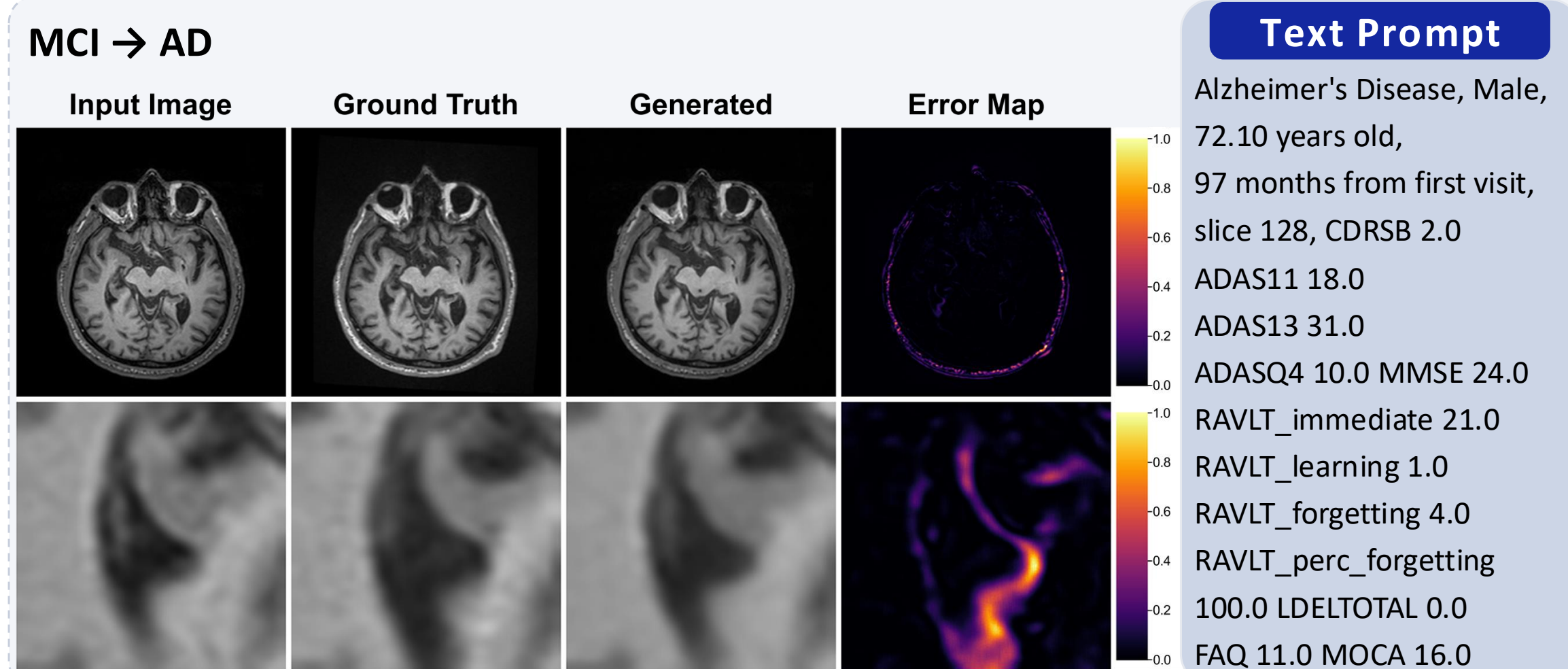
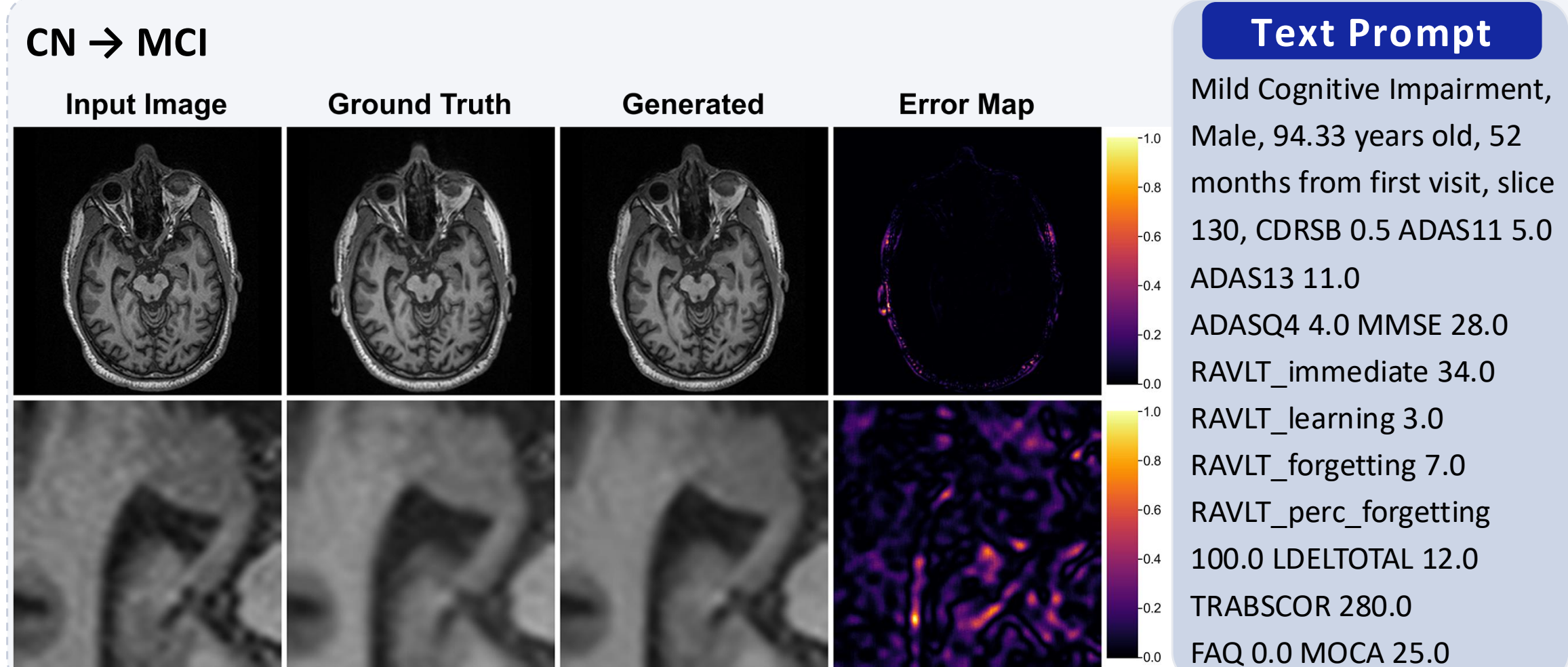
SSIM by Interval Δt	scale 0–1
0–12 mo	0.8975
12–24 mo	0.8437
24–36 mo	0.8380
≥ 36 mo	0.8305

6 Qualitative Results

Model Comparison with Absolute Error Maps



Progression cases



7 Conclusion

01 Clinical metadata–guided follow-up MRI synthesis with explicit interval control

02 Consistent SSIM and PSNR improvements over DiT and text-guided generative baselines

03 Anatomically plausible, subject-specific modeling of Alzheimer's disease progression